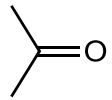
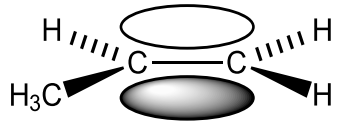
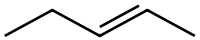
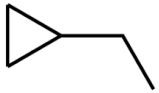
3(a)(i)		**1**
3(a)(iii)	elimination	**1**
3(a)(iv)		**1**
3(b)(i)	$(CH_3)_2CHOH + Na \rightarrow (CH_3)_2CHO^{(-)}Na^{(+)} + \frac{1}{2}H_2$	**1**
3(b)(ii)	**M1** correct dipole on $^{\delta+}C—Br^{\delta-}$ **AND** curly arrow from C—Br bond to Br **M2** intermediate  **M3** curly arrow from lone pair of $(CH_3)_2CHO^-$ to C^+	**3**

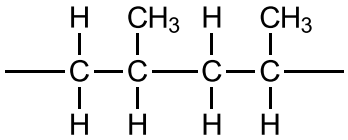
Examiner report missing
9701_w25_qp_24 Question 3

4(a)(i)	**M1** (molecules with the) same molecular formula / same number of atoms of each element **M2** different structural formulae/structures	**2**
4(a)(ii)	**M1**  **M2** restricted / lack of / no rotation about C=C	**2**
4(a)(iii)	electrophilic addition	**1**

Examiner report missing
9701_w25_qp_24 Question 4

3(a)	simplest /lowest (whole number / integer) ratio of atoms (of each element) in a compound / molecule	**1**
3(b)	two of:  **OR**  **OR** 	**2**
3(c)(i)	homolytic	**1**
3(c)(iii)	termination	**1**

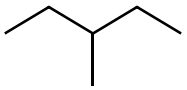
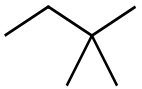
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9701_w25_qp_23 Question 3

4(a)(i)	reduction	**1**
4(a)(iv)	conc. H_2SO_4 **OR** conc. H_3PO_4 **OR** Al_2O_3 **AND** heat	**1**
4(b)(i)	addition	**1**
4(b)(ii)	**M1** (molecules that have the) same structural formula (and same molecular formula) **M2** but different 3D / spatial arrangement of atoms / groups	**2**
4(b)(iii)	 **M1** **M2** optical isomerism **M3** the methylated carbon has four different groups attached	**3**
4(c)(i)	2,3-dimethylbut-1-ene	**1**

Examiner report missing
9701_w25_qp_23 Question 4

3(a)	**M1** (family of molecules / compounds / series with) same general formula **OR** (family of molecules where) molecular formula differs (only) by (units of) $CH_{2(n)}$. **M2** similar chemical properties / reactions	**2**

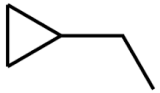
3(c)	cracking	**1**

3(d)(i)	**M1**  **M2** 	**2**

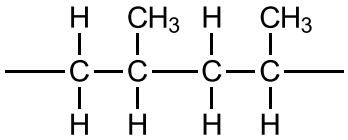
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9701_w25_qp_22 Question 3

4(a)(i)	free-radical substitution	**1**
4(a)(iii)	homolytic	**1**
4(a)(iv)	1,2-dichloropropane	**1**
4(a)(v)	**M1** (type of structural isomerism) positional **M2** (type of stereoisomerism) optical	**2**
4(c)(i)	orange to green	**1**
4(c)(ii)	$CH_3CH_2CH_2OH + 2[O] \rightarrow CH_3CH_2COOH + H_2O$	**1**
4(d)(i)	condensation	**1**

Examiner report missing
9701_w25_qp_22 Question 4

3(a)	simplest /lowest (whole number / integer) ratio of atoms (of each element) in a compound / molecule	**1**
3(b)	two of:  **OR**  **OR** 	**2**
3(c)(i)	homolytic	**1**
3(c)(iii)	termination	**1**

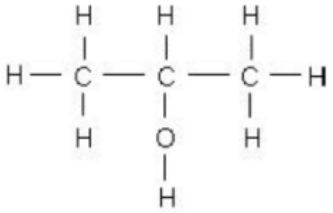
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9701_w25_qp_21 Question 3

4(a)(i)	reduction	**1**
4(a)(iv)	conc. H_2SO_4 **OR** conc. H_3PO_4 **OR** Al_2O_3 **AND** heat	**1**
4(b)(i)	addition	**1**
4(b)(ii)	**M1** (molecules that have the) same structural formula (and same molecular formula) **M2** but different 3D / spatial arrangement of atoms / groups	**2**
4(b)(iii)	 **M1** **M2** optical isomerism **M3** the methylated carbon has four different groups attached	**3**
4(c)(i)	2,3-dimethylbut-1-ene	**1**

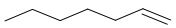
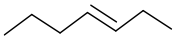
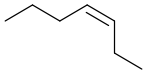
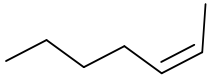
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9701_w25_qp_21 Question 4

2(a)(i)	non-linear	**1**
2(b)(ii)	oxidation number of S in SCl_2 = (+) 2 / (+) II oxidation number of S in H_2SO_3 = (+) 4 / (+) IV	**2**
2(b)(iii)	hydrolysis	**1**

Examiner report missing
9701_s25_qp_24 Question 2

4(a)	$C_5H_{10}O_2$	**1**
4(b)	propyl ethanoate	**1**
4(c)(i)	condensation	**1**
4(c)(ii)	$CH_3COOH + CH_3CH_2CH_2OH \rightarrow CH_3COOCH_2CH_2CH_3 + H_2O$	**1**
4(c)(iii)	propan-1-ol	**1**
4(d)(i)	$CH_3CN + HCl + 2H_2O \rightarrow CH_3COOH + NH_4Cl$	**1**
4(d)(ii)		**2**

Examiner report missing
9701_s25_qp_24 Question 4

5(a)	**M1**  **M2**  **OR** 	**2**
5(b)(i)	no rotation / restricted rotation of C=C / pi bond	**1**
5(b)(ii)	(**W** / it has two) different groups bonded to each C of the double bond	**1**
5(c)		**1**

Examiner report missing
9701_s25_qp_24 Question 5

6(a)(i)	simplest (whole number) ratio of atoms (of the elements) in a compound	**1**

6(a)(ii)	C : H **M1** $\frac{85.7}{12}$ or 7.14 : $\frac{14.3}{1}$ or 14.3 1 : 2 **M2** CH_2	**2**

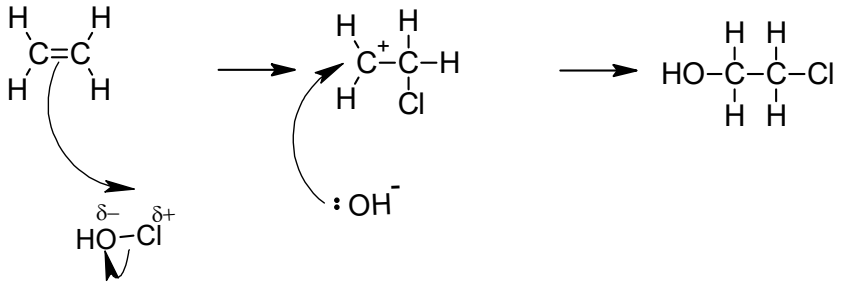
6(a)(iii)	alkene(s)	**1**

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9701_s25_qp_24 Question 6

5(a)(i)	C_2H_3O	**1**

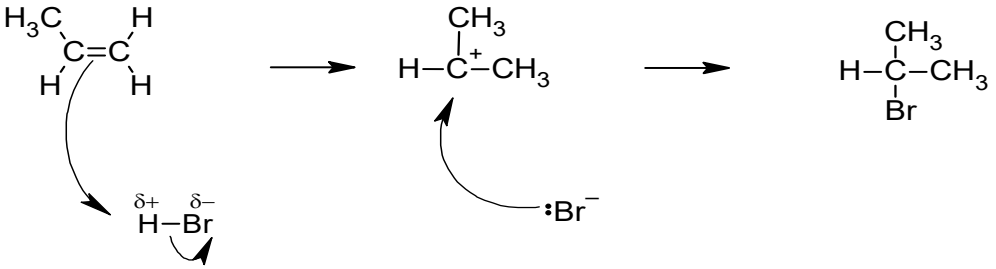
5(b)(i)	number of σ bonds = 14 number of π bonds = 2	**2**

Examiner report missing
9701_s25_qp_23 Question 5

6(a)(i)	two or more reactants / molecules react to make (only) one product	**1**
6(a)(iii)	water purification / killing bacteria or microbes	**1**
6(b)	 **M1** correct arrow for HO-Cl **M2** arrow from double bond of C=C to Cl **M3** correct intermediate **M4** arrow from lone pair on O of :OH- to the appropriate C	**4**
6(c)(i)	KCN/potassium cyanide in ethanol **AND** heat	**1**
6(c)(ii)	(acid) hydrolysis	**1**

Examiner report missing
9701_s25_qp_23 Question 6

3(a)(i)	two (or more) reactants / molecules react to make (only) one product	**1**

3(a)(ii)	 **M1** curly arrow from = of C=C to H (of HBr) **M2** curly arrow from bond of $H^{\delta+}$—$Br^{\delta-}$ to $Br^{\delta-}$ (dipole is essential) **M3** correct intermediate **M4** curly arrow from lone pair of Br^- to C^+ of intermediate drawn	**4**

Examiner report missing
9701_s25_qp_22 Question 3

4(a)(i)	$LiAlH_4$	**1**
4(a)(ii)	$CH_3CH_2CO_2H + 4[H] \rightarrow CH_3CH_2CH_2OH + H_2O$	**1**
4(b)(i)	substitution	**1**
4(b)(iii)	$CH_3CH_2CH_2CN + NaOH + H_2O \rightarrow CH_3CH_2CH_2CO_2Na + NH_3$ **OR** $CH_3CH_2CH_2CN + NaOH + 2H_2O \rightarrow CH_3CH_2CH_2CO_2Na + NH_4OH$ **M1** correct organic product (as molecule or anion) **M2** rest of equation is correct	**2**
4(b)(iv)	(alkaline) hydrolysis	**1**
4(c)(i)	sp^3	**1**
4(d)(i)	$(CH_3)_3COH$	**1**
4(d)(ii)	butan-1-ol **OR** (2) methylpropan-1-ol	**1**

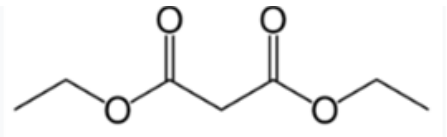
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9701_s25_qp_22 Question 4

5(a)	$C_3H_4O_3$	**1**

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9701_s25_qp_22 Question 5

4(a)	$C_3H_4O_3$	**1**

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9701_s25_qp_21 Question 4

5(a)	**M1** two ester links **M2** rest of structure correct 	**2**
5(b)	(acid) hydrolysis	**1**
5(d)(i)	**M1** NaOH(aq) **AND** heat **M2** acidified **AND** potassium dichromate(VI) **AND** heat under reflux	**2**
5(d)(ii)	oxidation	**1**
5(e)(i)	$LiAlH_4$	**1**
5(e)(ii)	**M1** correct unambiguous organic product **M2** correct balanced equation $HOOCCH_2COOH + 8[H] \rightarrow HOCH_2CH_2CH_2OH + 2H_2O$	**2**
5(e)(iii)	1,3-dichloropropane	**1**

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9701_s25_qp_21 Question 5

6(a)	carbonyl	**1**

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9701_s25_qp_21 Question 6

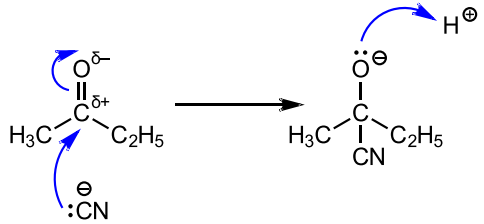
1(a)(i)	• (+)3 / III • –1 / I • (+)5 / V • –1 / I	**2**

1(c)(i)	contains a(/ one or more) C=C / double / multiple bonds	**1**

Examiner report missing
9701_m25_qp_22 Question 1

3(a)(i)	bromine: (dark / red-)brown liquid iodine: grey / black solid	**1**
3(d)(i)	instantaneous dipole—induced dipole / id—id / London / dispersion (forces) **AND** permanent dipole—permanent dipole / pd—pd (forces)	**1**
3(d)(ii)	electrophilic addition	**1**
3(d)(iii)		**2**
3(d)(iv)	position(al) / regio- (isomerism)	**1**

Examiner report missing
9701_m25_qp_22 Question 3

4(a)	<table><tr><td>acidified $K_2Cr_2O_7$(aq)</td><td>(solution) turns (from orange to) green</td><td>**J** and **L**</td></tr><tr><td>alkaline I_2(aq)</td><td>yellow precipitate forms</td><td>**K** and **L**</td></tr><tr><td>2,4-dinitrophenylhydrazine / 2,4-DNPH</td><td>orange precipitate forms</td><td>**J** and **K**</td></tr><tr><td>Na(s)</td><td>effervescence</td><td>**L** and **M**</td></tr></table>	**5**
4(b)(i)	 • curly arrow from lone pair on :CN to C($\delta+$) • curly arrow from C=O to O($\delta-$) **AND** correct dipole • correct intermediate • curly arrow from lone pair on O:$^-$ to H^+	**4**
4(b)(iii)	atom / carbon with four different atoms / (functional) groups attached / bonded	**1**

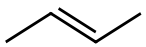
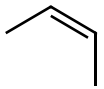
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9701_m25_qp_22 Question 4

1(a)(i)	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$	**1**

1(d)(i)	**M1** $\frac{79.5}{209.4} : \frac{20.5}{18.0} = 0.380 : 1.14 = 1:3$ OR $18x / (102.9 + 3(35.5) + 18x) = 0.205$ $x = 3$ OR $(102.9 + 3(35.5)) / (209.4 + 18x) = 0.795$ $x = 3$	**1**
	M2 $x = 3$	**1**

1(d)(ii)	molecules with the same structural formula and same molecular formula with different 3D / different spatial arrangement of atoms / groups	**1**

1(d)(iii)	a **single** product is made	**1**

1(d)(iv)		**1**
		**1**

Question 1

(a) (i) Most candidates were able to give a correct electronic configuration of: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$. A few candidates included the $3d^74s^2$ given in the question. This was not required in the answer.

(ii) This question was well answered with most drawings attempting to represent a sphere as a circle. A few diagrams showed the x, y and z axes drawn to represent a 3D shape.

(iii) Several candidates gave an answer of one, whilst many gave the correct answer.

(b) Most candidate answers were correct for the total number of protons and neutrons, but the total number of electron shells, for rhodium, were often incorrectly given as 4.

(c) (i) The definition was not well answered. Many candidates stated that it was the average mass of an isotope/weighted average mass of an isotope or the average mass of isotopes. Very few answers gave a complete definition.

(ii) Many candidates incorrectly used all three isotopes to calculate the A_r of iridium and divided by 100 or used the two isotopes of iridium but divided by 100 instead of by 50.

(d) (i) Many candidates were aware that to calculate an integer value of x, the water of crystallisation in $RhCl_3 \cdot xH_2O$, a mole ratio calculation was required. The ratio of H_2O to $RhCl_3$ was given/deduced as 20.5 per cent : 79.5 per cent by mass, from which a mole ratio of 0.38 : 1.14 can be calculated. Many candidates chose a slightly different route and used the expression $18x \div (209.4 + 18x) = 0.205$, to correctly calculate a vale of $x = 3$.

(ii) Many candidate answers mentioned that stereoisomers contained the same 'molecular formula' but omitted the fact that they also contained the same 'structural formula', which is the key feature for this definition. Several candidates correctly stated that the stereoisomers had 'different arrangements in space' but made no mention of what was arranged differently in space e.g. atoms/groups.

(iii) Many candidates failed to recognise that a single product would be formed and gave explanations about adding extra CH_2 units to the chain.

iv) The diagrams of the two stereoisomers of but-2-ene were often correct. Errors included confusion on the position of the C=C, in the chain, on one or both structures and incorrect groups or incorrect arrangements.

2(a)(i)	chlorine = yellow-green bromine = orange / brown / red iodine = silver grey / black	**1**
2(a)(iii)	green colour disappears	**1**
2(a)(iv)	H—Hal covalent bond strength **decreases** down the group	**1**

Question 2

(a) (i) The colours of the Group 17 elements, chlorine, bromine and iodine were not well known with common errors being 'yellow' for chlorine and 'purple' for iodine. Many candidate answers stated that the colours became darker down the group which was insufficient.

(ii) Most candidate answers correctly stated that the oxidising strength of the elements chlorine, bromine and iodine decreased down the group.

(iii) Candidate answers that were awarded credit stated that 'white fumes' would be observed although strictly this occurs only when moisture is present.

(iv) Many candidate answers linked the decreasing thermal stability of hydrogen halides to weaker intermolecular forces of attraction. Some candidates only mentioned bond length and did not link the decrease in thermal stability to the decrease in bond strength of the H-Hal bond.

(b) (i) Many candidate answers stated that it was an 'unpaired electron or a lone pair of electrons'. The requirement for a species to be specified, that contained 'one or more unpaired electrons', was often omitted.

(ii) This question was well answered.

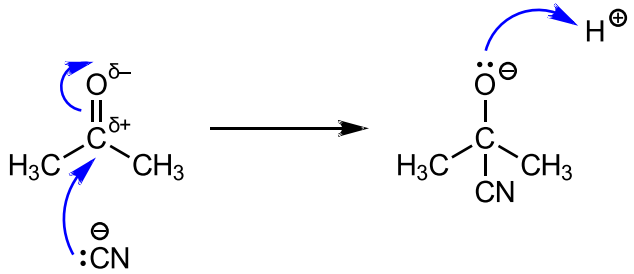
(iii) Few candidates were awarded full credit here. Incorrect responses showed that candidates were not clear what a propagation step involved and gave initiation or termination steps instead. Some included ions instead of radicals or gave H radical as an incorrect response.

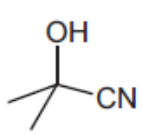
(c) (i) Many candidates recognised that NaOH was required to convert chlorine to NaClO, but the requirement for the NaOH to be in aqueous solution and the conditions to be 'cold' were regularly omitted.

(ii) The definition of disproportionation was well answered by many candidates.

(iii) This equation was well known.

(iv) This question was well answered by many candidates. Several candidates lost marks for giving additional incorrect observations for example: a brown gas of NO_2 is also formed. Those candidates who described how the Cl^- ion was formed, prior to its reaction with Ag^+ (in $AgNO_3$) lost credit by failing to appreciate that the $CHCl_3$ was initially hydrolysed to form Cl^-. Many stated that the Cl^- was contained in $CHCl_3$, and not a product from its hydrolysis.

4(a)(i)	 • curly arrow from lone pair on C of CN^- to carbonyl C • correct dipole on C=O • curly arrow from C=O bond to O • correct intermediate (the lone pair(s) not needed for this bullet point) • curly arrow from lone pair on —O to (source of) H^+	**1**
		1
		1

4(a)(ii)	A's central carbon atom is not bonded to four different atoms/groups of atoms **A** 	**1**

4(c)(i)	CH_3COCH_3 + **2**[H] → $CH_3CH(OH)CH_3$	**1**

4(c)(ii)	propan-2-ol	**1**

Question 4

(a) (i) The mechanism attracted many good answers. The common errors included: arrows from the negative charge (on the CN^- ion) instead of from the lone pair on the C atom, missing polarity, specifically missing the $\delta+$ on the H (of H-CN), and incorrect attack of the intermediate by the CN^- ion.

(ii) Many answers stated that there were two CH_3 groups on one C atom in A but did not go further to explain the significance of this in terms of the lack of optical isomerism in A. Credit was given for stating that no chiral centre was present or that the central C atom was not bonded to four different atoms/groups.

(b) Candidates often stated that the mineral acids HCl or H_2SO_4 were required for the hydrolysis reaction of a CN group to a carboxylic acid but missed the requirement for the acid to be dilute or aqueous.

(c) (i) The equation was frequently correct. A common error was the omission of 2(H), from the equation, or using just 2H, without the brackets.

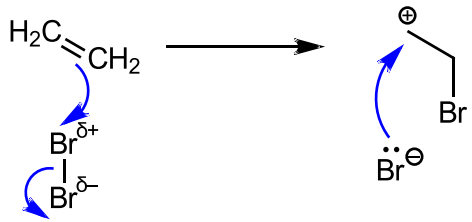
(ii) The name of compound C was well known.

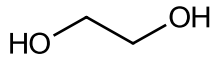
(d) Most candidate answers stated that an orange precipitate was formed when propanone was reacted with 2,4-DNPH.

(e) A common answer was that 'Fehling's reagent only reacts with aldehydes', with no reference to any chemistry in this reaction. Fehling's reagent behaves as an oxidising reagent and ketones are not easily oxidised, hence, there is no reaction between Fehling's reagent and propanone.

(f) (i) Many candidate answers stated that the absorptions at 2850 – 2950 cm^{-1} were attributable to a C-H bond. As all three compounds in the question contained groups containing C-H bonds these compounds were indistinguishable using these absorptions.

(ii) Many candidates correctly identified the absorption at 2200 – 2250cm^{-1}, which was unique to compound A, as it was the only compound containing a C≡N bond. As all three compounds, A, B and C contained an O-H bond, no credit was given for frequent mention of this absorption.

4(a)	 **pt 1** dipole on Br_2 **pt 2** curly arrow from C=C bond to a $Br^{\delta+}$ **pt 3** curly arrow from Br–Br bond to the other Br **pt 4** correct intermediate **pt 5** curly arrow from lone pair on Br^- to C^+ **5•** = 3 marks. **4•** = 2 marks. **3•** = 2 marks. **2•** = 1 marks. **1•** = 0 marks.	**3**

4(c)(i)		**1**
	bromoethene	**1**

4(d)(i)	structural / positional	**1**

4(d)(ii)	aldehyde	**1**

Question 4

(a) Well-rehearsed candidate answers included the accurate placement of curly arrows, dipoles and the correct intermediate. Weaker answers often had ambiguous placement of curly arrows, inappropriate use of partial charges, incorrect dipole on Br_2 and incorrect intermediates.

(b) Correct calculations, sometimes with an appropriate Hess' cycle in their working, were seen. Incorrect answers were calculated by those candidates that did not fully understand or use the definition of enthalpy change of formation when applying Hess's law.

(c) (i) The product **A** was correctly shown in many answers. Candidates need to show correct connectivity of the O-H group to the carbon backbone when structures are drawn.
The naming of **B** presented problems for some candidates. Incorrect names often described the incorrect position of Br on the carbon backbone or gave a name which described the wrong number of carbon atoms in the backbone.

(ii) Unambiguous structures were seen in a significant number of candidate answers.

(iii) This question proved to be extremely demanding. All the information provided in the details of Fig. 5.1 needed to be considered, including the three products as well as the reaction conditions.

(d) (i) Correct identification of the type of isomerism was given in a significant number of candidate answers. Chain isomerism was a common incorrect answer even though the two molecules were made up of only two carbon atoms.

(ii) Many candidate answers stated that **F** was a carbonyl compound; this term describes the functional group present in the molecule and is common to both homologous series; aldehyde and ketone.

(iii) Many candidate answers gave appropriate observations for the behaviour of **F** with 2,4-DNPH reagent and Tollen's reagent. A much smaller number of answers identified a positive reaction with appropriate observation for the reaction with alkaline I_2(aq).

(e) (i) This question proved to be demanding. The strongest candidates gave full explanations of their interpretation of the infrared spectrum by relating relevant wavenumber values at absorption troughs to appropriate bond types and used the molecular ion peak to deduce the molecular mass of **H**. Weaker responses quoted key wavenumber values at troughs in the infrared spectrum and restated that the molecular ion peak in the mass spectrum was at m/e = 60 without explaining what this information actually provides about structure **H**.

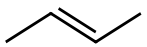
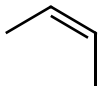
(ii) The correct role of reagent **G** was given in many candidate answers. Catalyst was a common incorrect answer.

1(a)(i)	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$	**1**

1(d)(i)	**M1** $\frac{79.5}{209.4} : \frac{20.5}{18.0} = 0.380 : 1.14 = 1:3$ OR $18x / (102.9 + 3(35.5) + 18x) = 0.205$ $x = 3$ OR $(102.9 + 3(35.5)) / (209.4 + 18x) = 0.795$ $x = 3$	**1**
	M2 $x = 3$	**1**

1(d)(ii)	molecules with the same structural formula and same molecular formula with different 3D / different spatial arrangement of atoms / groups	**1**

1(d)(iii)	a **single** product is made	**1**

1(d)(iv)		**1**
		**1**

Question 1

(a) (i) Most candidates were able to give a correct electronic configuration of: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$. A few candidates included the $3d^74s^2$ given in the question. This was not required in the answer.

(ii) This question was well answered with most drawings attempting to represent a sphere as a circle. A few diagrams showed the x, y and z axes drawn to represent a 3D shape.

(iii) Several candidates gave an answer of one, whilst many gave the correct answer.

(b) Most candidate answers were correct for the total number of protons and neutrons, but the total number of electron shells, for rhodium, were often incorrectly given as 4.

(c) (i) The definition was not well answered. Many candidates stated that it was the average mass of an isotope/weighted average mass of an isotope or the average mass of isotopes. Very few answers gave a complete definition.

(ii) Many candidates incorrectly used all three isotopes to calculate the A_r of iridium and divided by 100 or used the two isotopes of iridium but divided by 100 instead of by 50.

(d) (i) Many candidates were aware that to calculate an integer value of x, the water of crystallisation in $RhCl_3 \cdot xH_2O$, a mole ratio calculation was required. The ratio of H_2O to $RhCl_3$ was given/deduced as 20.5 per cent : 79.5 per cent by mass, from which a mole ratio of 0.38 : 1.14 can be calculated. Many candidates chose a slightly different route and used the expression $18x \div (209.4 + 18x) = 0.205$, to correctly calculate a vale of $x = 3$.

(ii) Many candidate answers mentioned that stereoisomers contained the same 'molecular formula' but omitted the fact that they also contained the same 'structural formula', which is the key feature for this definition. Several candidates correctly stated that the stereoisomers had 'different arrangements in space' but made no mention of what was arranged differently in space e.g. atoms/groups.

(iii) Many candidates failed to recognise that a single product would be formed and gave explanations about adding extra CH_2 units to the chain.

iv) The diagrams of the two stereoisomers of but-2-ene were often correct. Errors included confusion on the position of the C=C, in the chain, on one or both structures and incorrect groups or incorrect arrangements.

2(a)(i)	chlorine = yellow-green bromine = orange / brown / red iodine = silver grey / black	**1**

2(b)(i)	a species with one or more unpaired electrons	**1**

Question 2

(a) (i) The colours of the Group 17 elements, chlorine, bromine and iodine were not well known with common errors being 'yellow' for chlorine and 'purple' for iodine. Many candidate answers stated that the colours became darker down the group which was insufficient.

(ii) Most candidate answers correctly stated that the oxidising strength of the elements chlorine, bromine and iodine decreased down the group.

(iii) Candidate answers that were awarded credit stated that 'white fumes' would be observed although strictly this occurs only when moisture is present.

(iv) Many candidate answers linked the decreasing thermal stability of hydrogen halides to weaker intermolecular forces of attraction. Some candidates only mentioned bond length and did not link the decrease in thermal stability to the decrease in bond strength of the H-Hal bond.

(b) (i) Many candidate answers stated that it was an 'unpaired electron or a lone pair of electrons'. The requirement for a species to be specified, that contained 'one or more unpaired electrons', was often omitted.

(ii) This question was well answered.

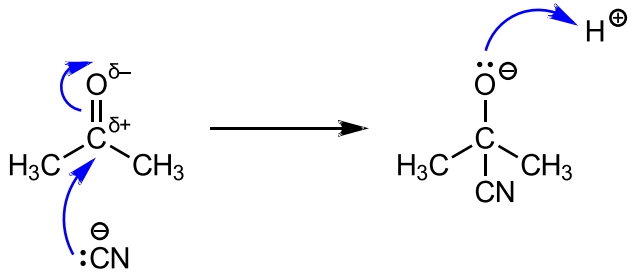
(iii) Few candidates were awarded full credit here. Incorrect responses showed that candidates were not clear what a propagation step involved and gave initiation or termination steps instead. Some included ions instead of radicals or gave H radical as an incorrect response.

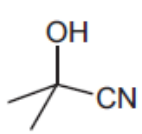
(c) (i) Many candidates recognised that NaOH was required to convert chlorine to NaClO, but the requirement for the NaOH to be in aqueous solution and the conditions to be 'cold' were regularly omitted.

(ii) The definition of disproportionation was well answered by many candidates.

(iii) This equation was well known.

(iv) This question was well answered by many candidates. Several candidates lost marks for giving additional incorrect observations for example: a brown gas of NO_2 is also formed. Those candidates who described how the Cl^- ion was formed, prior to its reaction with Ag^+ (in $AgNO_3$) lost credit by failing to appreciate that the $CHCl_3$ was initially hydrolysed to form Cl^-. Many stated that the Cl^- was contained in $CHCl_3$, and not a product from its hydrolysis.

4(a)(i)	 • curly arrow from lone pair on C of CN^- to carbonyl C • correct dipole on C=O • curly arrow from C=O bond to O • correct intermediate (the lone pair(s) not needed for this bullet point) • curly arrow from lone pair on —O to (source of) H^+	**1**
		1
		1

4(a)(ii)	A's central carbon atom is not bonded to four different atoms/groups of atoms **A** 	**1**

4(c)(i)	CH_3COCH_3 + **2**[H] → $CH_3CH(OH)CH_3$	**1**

4(c)(ii)	propan-2-ol	**1**

Question 4

(a) (i) The mechanism attracted many good answers. The common errors included: arrows from the negative charge (on the CN^- ion) instead of from the lone pair on the C atom, missing polarity, specifically missing the $\delta+$ on the H (of H-CN), and incorrect attack of the intermediate by the CN^- ion.

(ii) Many answers stated that there were two CH_3 groups on one C atom in A but did not go further to explain the significance of this in terms of the lack of optical isomerism in A. Credit was given for stating that no chiral centre was present or that the central C atom was not bonded to four different atoms/groups.

(b) Candidates often stated that the mineral acids HCl or H_2SO_4 were required for the hydrolysis reaction of a CN group to a carboxylic acid but missed the requirement for the acid to be dilute or aqueous.

(c) (i) The equation was frequently correct. A common error was the omission of 2(H), from the equation, or using just 2H, without the brackets.

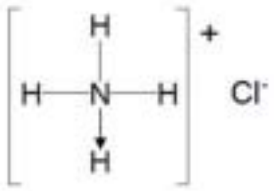
(ii) The name of compound C was well known.

(d) Most candidate answers stated that an orange precipitate was formed when propanone was reacted with 2,4-DNPH.

(e) A common answer was that 'Fehling's reagent only reacts with aldehydes', with no reference to any chemistry in this reaction. Fehling's reagent behaves as an oxidising reagent and ketones are not easily oxidised, hence, there is no reaction between Fehling's reagent and propanone.

(f) (i) Many candidate answers stated that the absorptions at 2850 – 2950 cm^{-1} were attributable to a C-H bond. As all three compounds in the question contained groups containing C-H bonds these compounds were indistinguishable using these absorptions.

(ii) Many candidates correctly identified the absorption at 2200 – 2250cm^{-1}, which was unique to compound A, as it was the only compound containing a C≡N bond. As all three compounds, A, B and C contained an O-H bond, no credit was given for frequent mention of this absorption.

4(a)	 **OR** Correct representation or label showing **M1** (ionic bond between) NH_4^+ and Cl^- **AND** 4 N—H covalent bonds OR 4 shared pairs of electrons in NH_4 **M2** 1 N—H coordinate bond OR arrow on bond/line in correct direction or only 1 bonding pair using the same symbol for electrons1	**2**
4(c)(i)	nucleophile	**1**
4(c)(ii)	OH^- (more reactive because) negatively charged species more attracted to halogenoalkane and / or $C^{\delta+}$ and / or intermediate and / or C^+ **OR** NH_3 (more reactive because) N electrons less tightly held **OR** N is more able to donate (lp) electrons (to make the covalent bond) **OR** N is less electronegative ORA	**1**

4(e)(i)	**M1** dipole and correct arrow on C—Br of 2-bromobutane **M2** lone pair on NH_3 attacking C(+) of the intermediate drawn **M3** curly arrow showing bond between N—H breaking and pointing towards the N	**3**

4(e)(iii)	butan-2-amine **OR** butyl-2-amine	**1**

Question 4

Many individual parts of this question were answered well. Parts **(a)**, **(c)(i)**, **(e)(ii)** and **(f)(ii)** were clearly the most challenging, often as candidates were required to apply several independent parts of their knowledge of the syllabus.

(a) Candidates were not strictly required to give dot-and-cross diagrams, though these and modified Lewis diagrams often produced the clearest responses. Many answers incorrectly showed a pentavalent nitrogen atom — a kind of NH_5 species, with little understanding of the NH_4^+ Cl^- superstructure. The dative bond needed to be clearly distinct from other covalent bonds and ionic charges needed to be given.

(b)(i) There were many correct answers to this question.

(ii) Different versions of this answer were offered, but a large number were a simple addition of NH_4^+ and OH^-, which did not gain credit.

(c)(i) This was well answered by candidates whose knowledge extended to terminology associated with electron-pair donors.

(ii) This was found to be the most challenging item on the paper which required candidates to assess either the basicity or nucleophilicity of each species. Opposing answers were therefore acceptable with enough reasoning.

(d) This was generally well answered by candidates, but the single-step nature of the S_N2 mechanism needed to be clearly evidenced by one transition state; the slower rate of this reaction by a higher energy of the TS relative to either produced in S_N1. Candidates frequently accessed the former part but often then erroneously showed some kind of catalysed reaction with a lower activation energy.

(e)(i) This question was not generally well answered, as candidates were not able to associate electron movement with the correct stage of the reaction and often were very imprecise with their positioning of curly arrows. Many answers had the arrowhead and tail reversed, with a lot of responses suggesting electron movement *from* extra H^+ species.

(ii) This question was reasonably well answered.

(iii) This question was generally well answered, although stray vowels or unspecific/incorrect numbering were observed. Candidates are reminded that the longest carbon chain is the important root of a molecule's name.

(f) (i) Candidates were able to provide a correct answer here by analogy of the 2-bromobutylium species on the left-hand side.

(ii) This was found to be a demanding question, although candidates who recognised the necessary reference to the stability of the intermediate were able to give a reasonably straightforward AO1 account for full credit.

5(a)	cold dilute $KMnO_4$ **OR** cold dilute potassium manganate(VII)	**1**
5(b)(i)	step 1 …..Br_2 (in the dark) step 2 …..NaOH(aq) + heat	**2**
5(b)(ii)	electrophilic addition	**1**
5(d)(i)	**W** =	**1**
5(d)(iii)	$C_8H_{12}O_4 + 8[H] \rightarrow C_8H_{16}O_2 + 2H_2O$	**1**

Question 5

This question tested organic reactions and spectroscopic methods using cyclohexene as a starting material. Many candidates seemed familiar with the chemistry, although the context of a cyclic molecule proved to be disconcerting for a few, who appeared to misinterpret the information given, or did not concentrate solely on the relevant functional groups.

(a) This was well answered by many, although differences appeared in the conditions to use with the $KMnO_4$ reagent (e.g. concentrated/dilute, alkaline/neutral/acidic).

(b)(i) Step 1 was familiar to many candidates. The hydrolysis of two functional groups in step 2 led many to think that H_2O_2 was being used, rather than two portions of NaOH.

(ii) In asking for a mechanism, 'electrophilic' was expected as well as 'addition'.

(c) Candidates showed some understanding in answering this question but often missed full credit by not following the rubric properly: the diagram needed labelling and each responsible bond (stretch) needed to be clearly identified. Many candidates only achieved half of this, or switched **F** and **G**.

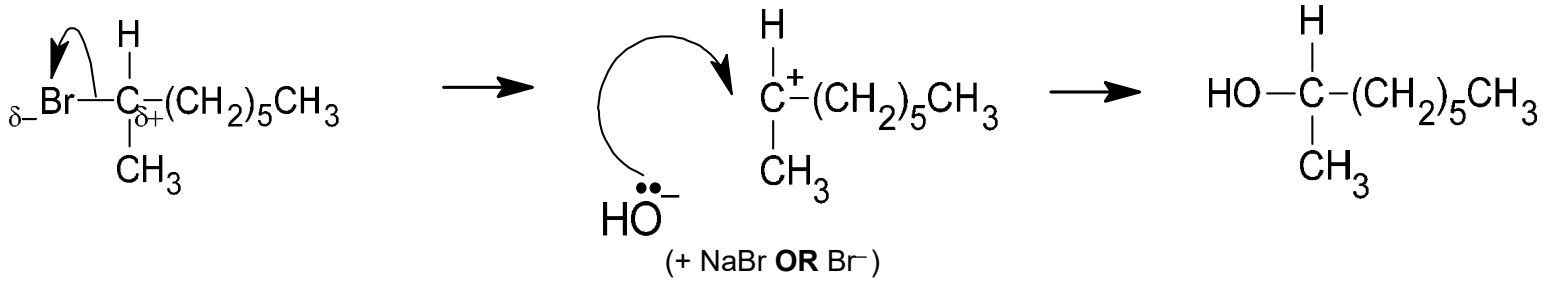
(d)(i) This was mostly well answered, although some candidates lost credit when opting to give fully displayed versions of the CN group, neglecting to show a triple C≡N bond or using the wrong connectivity (N attached to the ring).

(ii) This was answered well by many candidates. Others did not acknowledge that acidic conditions are used, creating an ammonium species rather than NH_3.

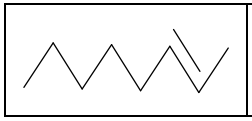
(iii) The formulae of the organic products were generally well indicated, though candidates often then misbalanced the equation with insufficient (H).

(iv) This was generally well answered by candidates. A few $NaBH_4$ responses were offered, to no credit: carboxylic acids need the stronger of the two hydride reagents on the syllabus.

4(a)	Three-dimensional stereoisomers of $CH_3(CH_2)_5CHBrCH_3$ **M1** correct representation of left-hand answer as 3d structure of one of the enantiomers of $CH_3(CH_2)_5CHBrCH_3$ **M2** correct representation of 3d structure of the second enantiomer of $CH_3(CH_2)_5CHBrCH_3$	**2**

4(b)	 (+ NaBr **OR** Br^-) **M1** correct dipole ($^{\delta+}C—Br^{\delta-}$) **AND** curly arrow from bond of C—Br to Br (or just beyond) **M2** correct intermediate **ALLOW** R for $–(CH_2)_5CH_3$ **M3** curly arrow from lone pair on OH^- to the C of intermediate (or suitable position where C–OH bond is made) *If $SN2$ mechanism shown:* Apply $SN2$ mechanism mark scheme **only** if mechanism shows OH / NaOH species attacking the reactant and no OH **M1** correct dipole ($^{\delta+}C—Br^{\delta-}$) AND curly arrow from bond of C—Br to Br of reactant (or just beyond) **NOT M2** **M3** curly arrow from lone pair on $:OH^-$ to the appropriate C of the reactant	**3**

4(d)(i)	elimination	**1**

4(e)(i)	 0 2 6 3 correct answers = 2 marks 2 correct answers = 1 mark 0 or 1 correct answers = 0 marks	**2**
4(e)(ii)	**M1** presence of a π bond **M2** limited rotation of C=C bond **M3 EITHER** two different groups on each / both double bonded carbon atoms **OR** two different atoms on each / both double bonded carbon atom **OR** one H / atom and one alkyl group on each / both double bonded carbon atom	**3**

Question 4

(a) Some excellent representations of the three-dimensional structures were seen; these included the approximate bond angles for these tetrahedral structures. Other diagrams showed confusion over the types of symbols used to represent bonds in the plane with those in and out of the plane. In order to remove ambiguity, candidates should be encouraged to use the convention described in the syllabus, under expected conventions for representing organic structures.

(b) Good knowledge of the S_N1 mechanism was described with precision in the best answers. Some confusion regarding the relative charges on the C–Br dipole and either no reference to a lone pair of electrons on O of OH^- or the location of the lone pair of electrons on the H of OH^- were common incorrect details. Some answers confused the S_N2 mechanism details with S_N1.

(c) The observations of these reactions were generally well known. Some answers did not include observations, as directed in the table heading, rather than attempted to predict the product of the reactions.

(d) (i) The name of this reaction was quite well known. Addition, substitution, dehydration and cracking were common incorrect answers.

(ii) The reagent NaOH was described in many answers. Confusion of the relevant conditions was seen in weaker responses.

(e) (i) Fully correct answers were seen in around half of responses. Sometimes confusion between two of the types of hybridisations was seen. The weakest answers demonstrated a lack of understanding of the different types of hybridisations which occur in carbon atoms in order to produce single, double and triple covalent bonds.

(ii) It was common for answers to contain features which lead to both types of stereoisomerism rather than focus on the stereoisomerism present in unbranched hydrocarbons. A relatively small proportion of answers showed good understanding of the features of an alkene required in order for it to exist as geometrical (cis/trans) isomers. A lack of precision in answers produced some ambiguity in responses, for example ‘no movement of a double bond’ rather than ‘no rotation of a double bond’.

Question 5

(a) (i) Those candidates with a good understanding of mass spectrometry realised that the peaks shown in Table 5.1 represented the molecular ion peak, M^+, and its $(M+1)^+$ peak and used the appropriate formula to calculate the relative abundance of the M+1 peak. Some weaker responses based their calculation on an incorrect formula.

(ii) The strongest responses engaged with the information about compound **W**. Weaker responses did not engage fully with the question details and answered in terms of structural formulae. Some formulae gave the fragments with no charges or negative charges.

(b) (i) An understanding that an O–H bond, due to a carboxyl functional group, produces a broad stretch within the range 2500–3000 cm^{-1} was seen in the best responses. The weakest responses did not engage with the details given for compound **W** and its reaction to produce **X**, and listed every potential functional group which might account for bond **A** and bond **B**, including amide and ester functional groups.

(ii) Of those candidates that correctly identified **B** as a C=O of a carbonyl group, some candidates ignored the reaction conditions stated in the question and represented **X** as an aldehyde rather than a ketone.

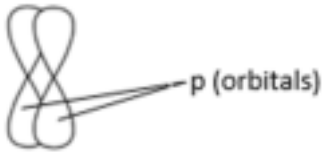
(c) (i) A significant proportion of candidates correctly named the functional group present in **Y**, based on the information provided.

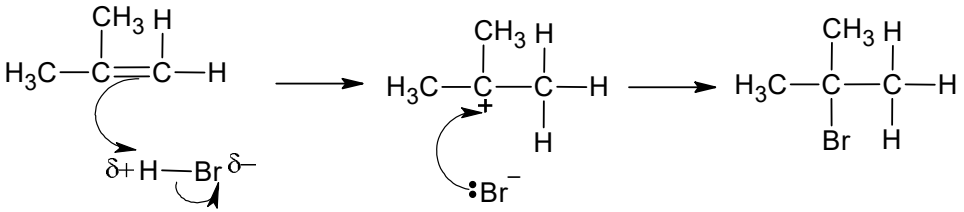
(ii) Many equations were shown with the correct formula of the organic product; not all equations were balanced. Identification of H_2O rather than H_2 as the other product was seen.

(iii) Those candidates who applied knowledge of the behaviour of different types of alcohol and carbonyl compounds correctly deduced the structure of **Y**. Confusion about the chemical reactions of alcohols and carbonyl compounds was seen in weaker responses.

5(a)	(electrostatic) attraction between nuclei of two atoms and shared pair of electrons	**1**

5(b)(i)	sp^2	**1**

5(b)(ii)	**M1** σ bond – diagram showing how σ bond is made from direct overlap of orbitals i.e.  OR **M2** π bond - diagram showing how π bond is made from sideways overlap of 2× p– orbitals i.e.  OR  OR  	**2**

5(c)(ii)	 **M1** arrow from = of C=C to H of H-Br **M2** correct dipole on H-Br **AND** arrow from H-Br bond to Br **M3** correct intermediate **AND** 2-bromopropane **M4** arrow from lone pair on Br^- to C^+ of their intermediate (with +ve or δ+ charge)	**4**

Question 5

(a) A minority of candidates gave the correct definition for a covalent bond. The most common errors included omitting a pair of electrons (being shared) and not mentioning that the attraction, for these electrons, was between the nuclei (of the two atoms).

(b) (i) The hybridisation of each carbon, in a C=C bond, was generally well answered.

(ii) Many candidates found this question difficult; some diagrams representing a σ bond were correct showing a linear overlap of two atomic orbitals or hybrids.

The diagrams for the formation of a π bond were often incorrectly drawn.
The most common mistakes included:

- not labelling p-orbitals, involved in the π bond formation
- drawing only the 'resultant' stage of π bond formation, showing the delocalised electrons above and below the nuclei, but without any labelling.

(c) (i) Credit was rarely awarded and most answers included a statement on how the σ and π bonds are formed by linear and sideways overlap of the relevant orbitals, which was then considered to render the π bond weaker than the σ bond.

The key idea is that during an electrophilic addition reaction the electrons in the π bond are involved as they are further from the carbon nuclei in the C=C bond and thus experience a weaker force of attraction from these nuclei.

(ii) Candidates who understood the reaction usually gained full credit. The main error involved the careless drawing of arrows, which often started from a C atom instead of from the middle of the π bond or from the Br, in a Br^- ion, instead of from a lone pair on the Br^- ion.

6(a)(i)	(molecules with the) same structural formula (and same molecular formula) with different arrangement of atoms / groups in space	**1**
6(a)(ii)	**M1** *number of stereoisomers* = 8 **M2** *reasoning*: 2 chiral centres **AND** 1 C=C (producing a cis-and trans or geometrical pair)	**2**
6(a)(iii)	$C_{13}H_{20}O_3$	**1**
6(a)(iv)	ester **AND** carbonyl / ketone **AND** C=C (bond) / alkene	**1**
6(b)(i)	reducing agent for C=O **AND** C=C	**1**

Question 6

(a) (i) The definition of stereoisomerism was one of the least well answered questions. The most common error was to state that 'molecules had the same molecular formula but different structural formula'. Few mentioned that it was the arrangement of the atoms or groups in space.

(ii) Very few answers gained full credit here. Some candidates identified a C=C double bond and two chiral centres and then incorrectly stated that this gave (2 × 3 =) 6 stereoisomers instead of using the correct expression 2^3 (= 8).

(iii) The molecular formula of **V** was answered quite well. There were no common errors.

(iv) Many candidates correctly identified an alkene and carbonyl group. The most common error was to omit the ester group and substitute it for an alcohol group or a carboxyl/carboxylic acid group.

(b) (i) The question required an identification of the role of reagent **T**. Candidates who stated that **T** behaved as a reducing agent gained full credit. Those candidates who ventured to explain the behaviour of **T** with each functional group generally understood that it reduced the carbonyl group in **V** to an alcohol group in **W**. The reaction of the C=C to from a saturated chain attracted a variety of incorrect answers ranging from acting as a catalyst to hydrolysing the alkene group.

(ii) This question was well answered.

(c) (i) Candidates found this very challenging. Many answers used the m/e values of 200 and 201, from the mass spectrum, instead of the relative abundance values of 100 and x, and attempted to average these values to determine the relative atomic mass.

Creditworthy answers deduced that **Q** contained 12 C atoms and then used this value of *n* in the expression $n = \dfrac{100 \times \text{abundance of M} + 1 \text{ ion}}{1.1 \times \text{abundance of M} + \text{ion}}$

to calculate the abundance of the (M + 1) ion which is equivalent to the value of x.

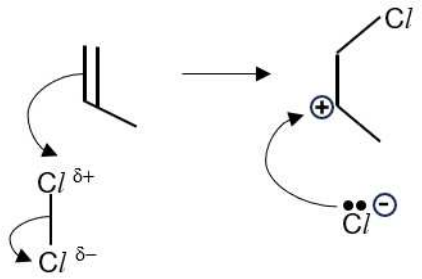
(ii) This question was answered correctly by many candidates with the most common error being to omit 'precipitate' with the orange colour for the observation with 2,4-DNPH with **Y**.

(iii) The majority of candidates did not recognise that the volume of hydrogen gas was collected at s.t.p., in the question, which, from the Table of data, is 22.4 $dm^3 mol^{-1}$.

Various methods were used to calculate the moles of gas produced to include the ideal gas equation and 24.0 $dm^3 mol^{-1}$.

Those candidates who managed to calculate the number of moles of gas then needed to appreciate that the ratio of hydroxyl groups to hydrogen gas was 2 : 1. The award of full credit was rare.

(iv) Many candidates gave correct absorption ranges for one of the two carbonyl absorptions in **Y**, for the ketone and ester functional groups. Several answers gave a range of absorptions for a carboxyl group in **Q** as a difference in the infrared spectrum, having not appreciated that **Q** contains **only** hydroxyl groups, mentioned in the stem of **(c)(ii)**.

2(a)	<table><tr><td>formula of chloride</td><td>● NaCl</td><td>● $MgCl_2$</td><td>● $SiCl_4$</td></tr><tr><td>structure</td><td>giant</td><td>● giant</td><td>● simple</td></tr><tr><td>bonding of chloride</td><td>● ionic</td><td>● ionic</td><td>covalent</td></tr><tr><td>pH of solution formed</td><td>● 7</td><td>6.2</td><td>● 1–4</td></tr></table>	**3**
2(c)(i)	● trigonal bipyramidal ● octahedral	**2**
2(c)(ii)	H_2O / water	**1**
	hydrolysis	**1**
2(d)(i)	 ● curly arrow from C=C bond to $Cl^{(\delta+)}$	**1**
	● correct dipole on Cl_2 **AND** curly arrow from Cl–Cl bond to $Cl^{\delta-}$	**1**
	● correct carbocation intermediate	**1**
	● curly arrow from lone pair on Cl^- to C^+	**1**

Question 2

This was a holistic question based around the chemistry of chlorine and chlorides. Candidates are recommended to give brief answers to one-mark questions and to draw diagrams clearly.

(a) Common errors were found in the formulae of the chlorides, the structure of $SiCl_4$ and in the pH of a $NaCl$ solution (neutral so pH 7; candidates often seemed to think that: $NaCl + H_2O \rightarrow HCl + NaOH$).

(b) (i) Many variant and phonetic spellings of 'disproportionation' were observed.

(ii) There was an issue with this question in that water was not included as one of the products of the reaction, so full marks were awarded to all candidates to make sure no candidates were disadvantaged. This has been corrected in the published version of the paper.

(iii) This equation was rarely constructed correctly: where more valid attempts were made, common errors were either that $NaClO$ was given as the oxidation product or that the correct stoichiometric coefficients were omitted.

(iv) This was answered well by most candidates, although a number offered effervescence as an observation.

(c) (i) Octahedral was the most common answer for PCl_6^-, although commonly hexagonal or octagonal were seen (and not credited). A broad range of answers were given for PCl_4^+, with 'trigonal pyramidal' (missing the bi- prefix) and 'bipyramidal' (missing the base shape) among them.

(ii) **J** was often correctly identified as H_2O/water. Hydration or displacement were often given instead of hydrolysis or substitution for the type of reaction.

(d) (i) Some very loosely drawn diagrams were unable to be given full credit: curly arrows should come from a clear origin (the middle of a bond, or from pair of electrons, rather than a negative charge) and, in cases such as this, the arrowhead needs to have both 'blades' and point to a specific atom or the middle of a bond.

Many candidates also confused partial charges/dipoles with full formal charges.

(ii) Candidates were required to reference heating the ethanolic NaOH to gain full credit.

(iii) Many answers retained the C=C or arranged the 3C monomer linearly with no branching. These were both common and recurring errors in questions of this type.

4(a)	<table><tr><td>molecule</td><td>**P**</td><td>**Q**</td><td>**R**</td><td>**S**</td></tr><tr><td>geometrical</td><td>✓</td><td>(×)</td><td>(×)</td><td>(×)</td></tr><tr><td>optical</td><td>(×)</td><td>✓</td><td>✓</td><td>✓</td></tr></table>	**1**
	<table><tr><td>molecule</td><td>**P**</td><td>**Q**</td><td>**R**</td><td>**S**</td></tr><tr><td>geometrical</td><td>✓</td><td>(×)</td><td>(×)</td><td>(×)</td></tr><tr><td>optical</td><td>(×)</td><td>✓</td><td>✓</td><td>✓</td></tr></table> $CH_3CHBrCH_2CH_3$	**1**
4(b)(i)	<table><tr><td>molecule</td><td>**P**</td><td>**Q**</td><td>**R**</td><td>**S**</td></tr></table>	**1**
4(b)(ii)	<table><tr><td>geometrical</td><td>✓</td><td>(×)</td><td>(×)</td><td>(×)</td></tr><tr><td>optical</td><td>(×)</td><td>✓</td><td>✓</td><td>✓</td></tr></table> $CH_3CHBrCH_2CH_3$ nucleophilic substitution	**1**
4(c)(i)	n = 100 / 1.1 × 0.55 / 7.2	**1**
	= 6.94 (so 7 carbon atoms)	**1**
4(c)(ii)	FG = ester / (R)COOR(')	**1**
	(absorption at) 1040–1300 cm^{-1} indicates C—O (stretch) **AND** (absorption at) 1710–1750 cm^{-1} indicates (C=O stretch)	**1**

Question 4

This an organic and spectroscopic question. Candidates are advised to be clear in matters of structure, both when representing molecules diagrammatically and when referring to features within a molecule, such as those identified by different types of spectra.

(a) This was answered well by many candidates. There were incidences of ambiguous answers, such as a 'crossed tick' (✓✗). Credit cannot be awarded if an answer is ambiguous.

(b) (i) Unambiguous structural formulae only were accepted, therefore molecular formulae were discounted and those in which the connectivity was questionable.

(ii) There were many correct answers to this question.

(iii) This proved to be one of the most challenging questions on the paper, requiring the balanced inclusion of (2)H_2O and H^+ on the left-hand side, and NH_4^+ on the right. Salts from viable reactions of specific acids were accepted.

(c) (i) Candidates who understood the $M^+ : [M+1]^+$ ratio were generally able to gain full credit here. Occasional arithmetical errors were seen.

(ii) Many candidates gave 'ester' as a correct answer but were then unable to corroborate this with sufficient and necessary information from the spectrum. Reference to an absorption outside of the C=O ester region contradicted an answer; equally, an absorption within the 1710–1750 cm^{-1} window was not solely sufficient, because of overlap with other absorptions. Corroboration could be given with reference to the C–O stretch (and the lack of O–H stretch).

C–H stretches are always ignored in these situations.

Absorptions must be accompanied by with assignment to an identified bond.

(iii) Propan-1-ol was a common incorrect answer for **T**. OH– is discouraged as a representation of an alcohol group. Many candidates did not draw the branched C5 acid correctly: some structures were unbranched; others had too many or too few H atoms.

3(a)	compound / molecule / substance that oxidises another / is reduced	**1**

3(e)(iii)	molecules with the same molecular formula but different structural formulae	**1**

Question 3

(a) The definition of an oxidising agent, as a species that oxidises another species, and is reduced in the process by the gain of electrons, was overall generally well answered.

(b) This question required the completion of an energy diagram/energy profile to represent the catalytic effect of MnO_2(s) on the decomposition of $KClO_3$(s). This entailed drawing two labelled curves representing the uncatalysed and the catalysed profiles.

Many answers were awarded full credit. Common errors included unlabelled diagrams or profiles with only one curve.

(c) The term *disproportionation* is used to describe the oxidation and reduction of a single element, in the same chemical reaction. Here the reaction involved $KClO_3$ being converted to $KClO_4$ and KCl, in which the Cl is both oxidised and reduced in this reaction.

Many chose the correct element; there were, however, several answers that identified K or $KClO_3$ as the species undergoing simultaneous redox.

(d) (i) PV=nRT was successfully used by many candidates to calculate the volume of 1 mole of gas, in a reaction of glucose with molten $KClO_3$, to give the correct answer of 0.0524 m^3, to 3 significant figures.

$$4KClO_3 + C_6H_{12}O_6 \rightarrow 6CO_2 + 6H_2O + 4KCl$$

Several candidates did not appear to read the question carefully and gave answers with an incorrect number of sig. figs and/or incorrect rounding of their answers.

(ii) This question required the calculation of the moles of glucose, (5/180) and to use this value, together with their answer from 3**(d)(i)**, to calculate total volume of the gaseous CO_2 and H_2O in the equation.

Many candidates were able to correctly calculate the moles of glucose but then failed to multiply this value by 12 to determine the total moles of CO_2 and H_2O produced.

Several answers then continued to repeat the calculation in 3**(d)(i)** instead of using their answer from 3**(d)(i)**.

(e) (i) A high proportion of candidates correctly identified the number of primary, secondary and tertiary alcohol groups, in glucose, as 1, 4 and zero, respectively.

(ii) This question required observations for three tests.

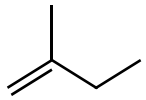
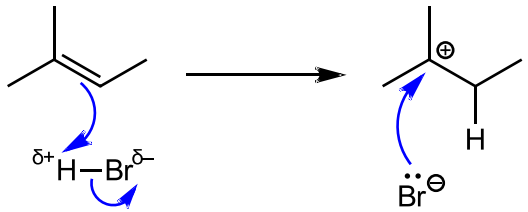
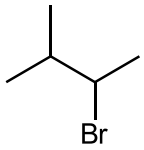
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Glucose + warm, alkaline I_2(aq) was incorrectly stated as a yellow precipitate by many candidates. There is no reaction with glucose, so the solution remains colourless.

(iii) Several candidates were able to correctly define *structural isomers* as 'molecules with the same molecular formula but different structural formulae'.

4(a)(i)	3-methylbut-1-ene	**1**
4(a)(ii)	one end / one C of the C=C bond has the same groups attached	**1**
4(a)(iii)		**1**
4(c)(i)	 correct dipole on H-Br AND curly arrow from C=C bond to $H^{\sigma+}$ **AND** curly arrow from H—Br bond to $Br^{\sigma-}$	**1**
	correct (Markovnikoff) intermediate	**1**
	curly arrow from lone pair on Br^- to C^+ of intermediate	**1**
4(c)(ii)		**1**

Question 4

(a) (i) Most candidates correctly gave the systematic name of the alkene **D**, with a molecular formula of C_5H_{10}, as **3-methylbut-1-ene**.

(ii) Few candidates gave a correct explanation of why the alkenes, **C** and **D**, did not exhibit geometric isomerism.

In **C**, one C (in the C=C bond) has two methyl groups and in **D**, one C (in the C=C) has two hydrogen atoms; neither will exhibit geometrical isomerism.

(iii) Many candidates gave pent-2-ene, or **C** or **D** as examples of a positional isomer.

(iv) When **D** is reacted with $H_2(g)$, 2-methylbutane, $CH_3CH(CH_3)CH_2CH_3$, is formed. Many candidates omitted the brackets surrounding the side-chain methyl group, on the CH group.

(v) Many candidates gave a monomer containing only two methyl groups or structures that still contained a double bond.

(b) The question contained information regarding the relative intensities of the M and (M + 1) ions, from a mass spectrum of **C**, which showed a molecular ion peak at 70.

Many candidates correctly calculated that C contained 5C atoms but then failed to use this information, in conjunction with the molecular ion peak, 70, to confirm that the molecular formula of C was C_5H_{10}.

(c) (i) The mechanism for the addition of HBr to **C,** 2-methylbut-2-ene, was quite well answered. Common errors included full charges on the H-Br, instead of a dipole, curly arrows, from C=C to $H^{\delta+}$ of HBr, arrows starting at one end of the C=C bond and intermediates with both a –ve charge and a partial negative charge.

(ii) This question was well answered with a common error being the use of C_3H_7, instead of the unambiguous expanded version, $(CH_3)_2CH–$.

(iii) This question required an explanation as to why different major products were formed when HBr reacts separately with C and D.

Many candidates appreciated that methyl/alkyl groups are responsible for a positive inductive effect and that the more alkyl groups, surrounding the carbocation, the more stable is the carbocation intermediate. However, the application of this concept to the intermediates of C and D was not well recognised.

Common errors were statements that the tertiary halogenoalkane was more stable, in C, and the secondary halogenoalkane was more stable, in D, rather than their respective intermediate carbocations.

Many candidates quoted Markovnikov's rule without any further explanation.

(d) *Please note that due to an issue with diagram labelling in this question, full marks have been awarded to all candidates to make sure that no candidates were disadvantaged. This has been corrected in the published version of the paper.*

The question required completion of a table of reagents and conditions for a reaction scheme.

The formation of alcohol, **F**, from a halogenoalkane, **E**, required the use of NaOH(aq).

Cleavage of the double bond, in **C**, to form a carboxylic acid, **G**, and propanone, was carried out using hot acidified <u>concentrated</u> $KMnO_4$.

The formation of ester, **H**, from **F** and **G**, needed <u>concentrated</u> H_2SO_4 (as a catalyst).

3(a)(i)	$4P / P_4 + 5O_2 \rightarrow P_4O_{10}$	**1**
	$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$	**1**
3(c)(i)	H_2O / steam	**1**
3(c)(iii)	condensation	**1**

Question 3

(a) (i) Fully balanced equations were seen by more able candidates.

(ii) This definition was described accurately in most candidates answers. A common misconception was seen in weaker responses that described a weak acid as one that *'did not completely dissolve'*.

(b) (i) Many candidates gave a definition of the enthalpy change which occurs when 'one mole of product is formed from its elements'. It was less common for the elements to be described as 'in their standard states'.

(ii) More able candidates gave excellent and logical workings that contained clearly labelled energy cycles. It was common to see an expression for the different bond energies added or subtracted from each other, in a variety of possible incorrect combinations, with no attempt to show how these expressions were deduced. The stoichiometry of the equation was ignored by some.

(iii) Excellent answers were seen which identified phosphorus in H_3PO_3 as both oxidised and reduced and linked the process of oxidation and reduction to the relevant changes in oxidation number for the three phosphorus containing species. Weaker responses were less specific, for example '*H_3PO_3 is oxidised and reduced'*. Oxidation numbers were sometimes described incorrectly.

(c) (i) Some candidates identified **A** as steam or $H_2O(g)$, other candidates were less precise and identified **A** as water. A common incorrect answer was NaOH.

(ii) Many unambiguous representations of 2-bromopropane were seen. Occasionally the central carbon atom was drawn with a valency of 3.

(iii) Description of this esterification reaction as a condensation reaction was relatively well known. Incorrect answers included hydrolysis and oxidation.

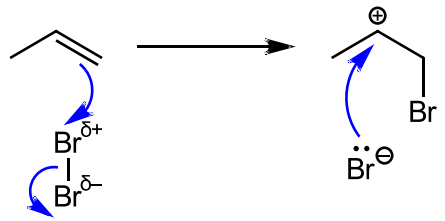
(iv) This question proved to be demanding. Candidates were expected to ignore all bonds which were common to both reactant and product. While the appropriate bond, C=O, was identified in many answers C-O and C-H were also seen. Most candidates only identified a range of wavenumbers where a change in absorption occurs rather than described if the absorption appeared, disappeared or moved.

(d) (i) More able candidates deduced a formula for the correct organic product in the equation. The inclusion of H_2O as the additional product was less common, even when included the equation was not always balanced.

(ii) The appropriate formula to deduce the number of carbon atoms present in **T** was not well known and many confused calculations were seen. It was common for candidates to give a structural formula for **T** rather than the required molecular formula.

4(a)(i)	(it has molecules that) contain a chiral carbon / centre **OR** are non-superimposable mirror images	**1**

4(a)(ii)	2-hydroxypropanoic acid	**1**

4(b)(i)	 curly arrow from C=C bond to $Br^{\delta+}$ **AND** curly arrow from $Br^{\delta+}$— $Br^{\delta-}$ bond to $Br^{\delta-}$ **AND** $Br^{\delta+}$ closest to the double bond	**1**
	correct intermediate	**1**
	curly arrow from lone pair on Br^- to $C^{(+)}$ of the intermediate	**1**

Question 4

(a) (i) Some candidates gave clear explanations of the presence of a chiral carbon in the lactic acid structure. A few candidates stated that optical isomers had chiral centres or described the behaviour of an optical isomer in the presence of polarised light.

(ii) Many candidates were unable to give the systematic name of lactic acid. 'Carboxylic acid' was seen frequently.

(iii) It was common for diagrams to show **more** than one water molecule or **multiple** hydrogen bonds. Candidates should be advised to respond as directed by the question. It was quite common for answers, showing more than one hydrogen bond, to contradict a correct representation.

(b) (i) Some clear and precise reaction mechanisms were shown with partial charges assigned to relevant atoms and unambiguous representation of curly arrows to show bonds broken and bonds made. When drawing mechanisms of this type candidates should be encouraged to consider what the curly arrows represent and whether their representation is clear, use partial charges correctly and discriminately and check that the number of carbon atoms on the intermediate is correct.

(ii) The equations described sometimes contained only pyruvic acid as a product or identified H_2 rather than H_2O as a product.

(iii) This question was demanding. Many candidates confused the reagents and conditions required to convert halogenoalkanes to nitriles with those required to convert ethanal to a hydroxynitrile. Deduction of the reagents and conditions used in the second step of both synthesis from propene and ethanal proved demanding for most candidates. Oxidising agents were more commonly seen for the third steps, however, some candidates represented the oxidising agents with incorrect formulae or with no reference to acid conditions.

3(a)	compound / molecule / substance that oxidises another / is reduced	**1**

3(e)(iii)	molecules with the same molecular formula but different structural formulae	**1**

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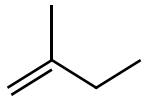
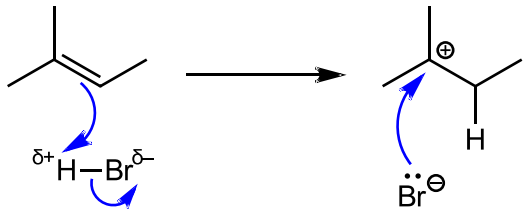
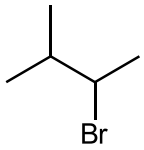
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The formation of ester, **H**, from **F** and **G**, needed concentrated H_2SO_4 (as a catalyst).

5(a)	C_3H_7O	**1**

5(b)(i)	$C_6H_{10}O_3 + [6H] \rightarrow C_6H_{14}O_2 + H_2O$	**1**

5(c)	<table><tr><td>type of hybridisation</td><td>sp^2</td><td>sp^3</td></tr><tr><td>number of carbon atoms in **X**</td><td>2</td><td>4</td></tr></table>	**2**

Question 5

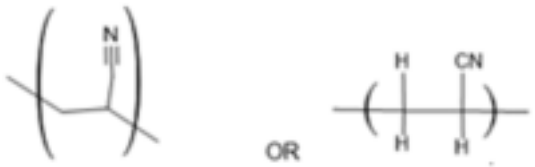
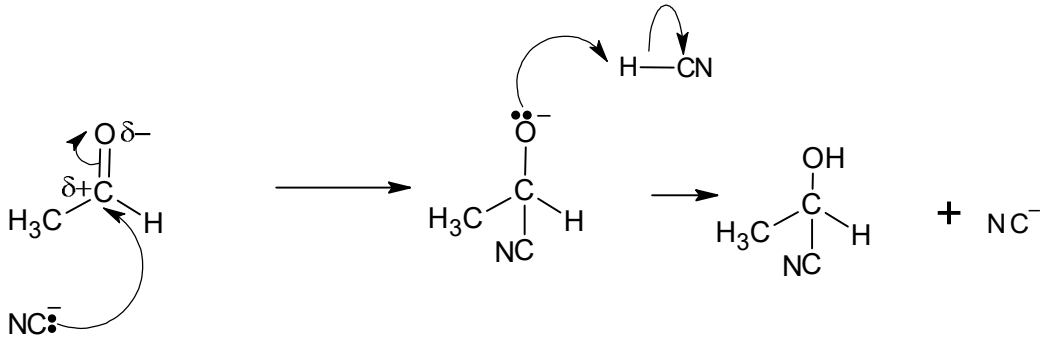
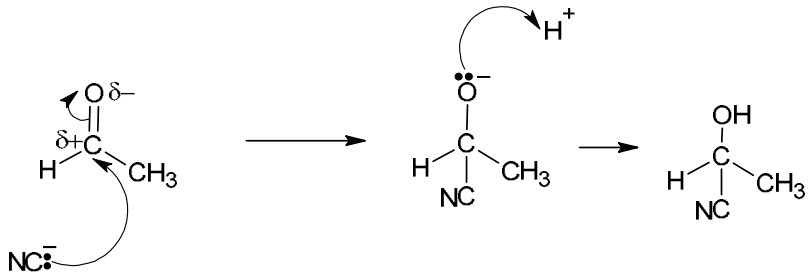
(a) Some candidates deduced the correct empirical formula of C_3H_7O from a molecular formula of $C_6H_{14}O_2$.

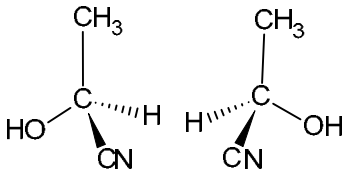
(b) (i) Very few candidates used molecular formula, as required in the question, to construct an equation for the reduction reaction.

(ii) Many answers correctly stated that a suitable reducing agent for the reaction is $LiAlH_4$.

(c) Some good answers were seen.

(d) Many candidates correctly stated that **X**, the ketoacid, would react with all three reagents to give effervescence, an orange precipitate and yellow precipitate respectively. Fewer candidates recognised that the diol, **Y**, would only react with the alkaline $I_2(aq)$ as it contained a $CH_3CH(OH)$ moiety.

6(a)	 OR	**1**
6(b)(i)	nucleophilic addition	**1**
6(b)(ii)	 **OR**  **M1** correct dipole on C=O **AND** curly arrow from C=O bond to O **AND** from lone pair of electrons on C of :CN^- **M2** correct intermediate (charge on O or overall) **M3** curly arrow from lone pair of electrons on O to H^+ **AND** correct organic product (+ CN^-)	**3**
6(c)(i)	(molecules / compounds with the) same structural formula (and same molecular formula) with different arrangement of atoms / groups in space	**1**

6(c)(ii)	**M1** restricted rotation of C=C (bond) **M2** each C of the double bond has two different groups / two different atoms (attached)	**2**
6(d)	 A pair of molecules of 2-hydroxypropanenitrile with bonds represented in 3-d **AND** the pair of molecules are non-superimposable images	**1**
6(e)	$CH_3CH_2CN + 2H_2 \rightarrow CH_3CH_2CH_2NH_2$	**1**
6(f)(i)	substitution	**1**

Question 6

(a) The single repeat unit derived from the monomer, CH_2=CHCN, by addition polymerisation was quite well answered.

(b) (i) Many candidates recognised that the mechanism for step 1 was nucleophilic addition.

(ii) Overall, the mechanism for step 1 was generally not answered well. The polarity on the C=O bond was often absent and the arrow from the CN^- to the C (in the C=O bond) did not start from the lone pair on the C of the CN^- ion. Several answers contained an incorrect intermediate with a positive charge which was then attacked by the CN^- nucleophile and many answers incorrectly showed an arrow from an H^+ ion to an O^- (in the intermediate).

(iii) Some good answers were seen.

(iv) The absorptions in the infrared spectrum of CH_2=CHCN were often correctly identified as C≡N and C=C. Incorrect answers mainly contained an identification of the functional groups rather than the bond or identifying the absorption (annotated as T on the IR spectrum) as that of a C=O group.

(c) (i) Very few candidates gave an accurate description.

(ii) A number of answers were awarded some credit. Few candidates were able to give a full description.

(d) Several candidates gave a correct pair of molecules that were non-superimposable images of $CH_3CH(OH)CN$. A common error was diagrams containing the same type of bond at 180° to each other.

(e) The equation for the reduction of propanenitrile to propylamine using hydrogen gas was not well known. Few candidates used the correct stoichiometry for hydrogen.

(f) (i) It was well known that the name of the type of reaction in step 1 is substitution.

(ii) Many candidates gained credit for ammonia. Many omitted heat or the solvent as conditions.

3(a)(i)	KCN in ethanol + heat	**1**
3(a)(ii)	propanenitrile	**1**
3(a)(iii)	(acid) hydrolysis	**1**
3(c)	<table><tr><td>type of bond in **X**</td><td>number of bonds in **X**</td></tr><tr><td>sigma σ</td><td>8</td></tr><tr><td>pi π</td><td>2</td></tr></table>	**2**

Question 3

(a) (i) Some candidates incorrectly described the reagent and conditions required to convert carbonyl compounds to hydroxynitriles, rather than halogenoalkanes to nitriles.

(ii) The systematic name for **X** was not always used, e.g. propionitrile or ethyl cyanide. Omission of the 'e' in propanenitrile was often seen.

(iii) It was generally well known that this type of reaction is a hydrolysis reaction.

(iv) Many candidates correctly produced equations representing the production of hydrogen gas. The formula of the calcium salt was frequently given in terms of calcium with a valency of one.

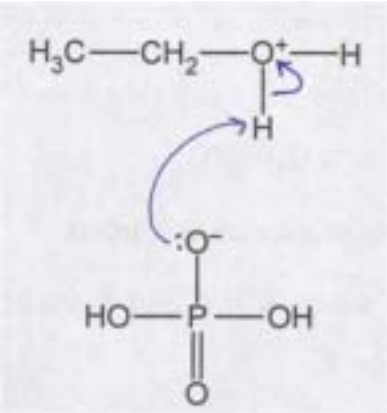
(b) Many candidates knew that acidic conditions were required. Stronger candidates appreciated that the essential condition for this reaction was 'heat under reflux' to ensure complete oxidation to the carboxylic acid.

(c) Most candidates described the correct number of pi bonds present in **X**; the correct number of sigma bonds was seen less frequently. Seven, nine and ten sigma bonds were common incorrect answers.

4(a)	Method 1 • identification of number and type of bonds when all are broken 2O-H + 4C-H + C=C **OR** (2 × 460) + (4 × 410) + 610 • Identification of number and type of bonds when all are made O-H + 5C-H + C-C + C-O **OR** (460 + (5 × 410) + 350 + 360) • calculation of energy to break bonds – energy to make bonds 3170 – 3220 = – 50 (kJ mol^{-1}) Two correct for one mark, three correct for two marks. **OR** Method 2 • identification of number and type of relevant bonds broken (including the new C-H only) 2O-H + C=C **OR** (2 × 460) + 610 • identification of number and type of relevant bonds made (including the new C-H only) O-H + C-H + C-C + C-O **OR** (460 + 410 + 350 + 360) • calculation using energy to break bonds – energy to make bonds 1530 – 1580 = –50 (kJ mol^{-1}) Two correct for one mark, three correct for two marks.	**2**

4(c)(i)	acid **AND** it donates a proton / H^+	**1**

4(c)(ii)	$(CH_3CH_2)^+$ OR $CH_3C^+H_2$ **AND** accepts a pair of electrons	**1**

4(c)(iii)	 **M1** arrow from bond of one of the O-H to O of the cation **M2** arrow from lone pair on O of the anion to between H of H-O	**2**

Question 4

(a) Stronger candidates showed clear logical working and selection of appropriate data. Common errors included mis-reading the table, using an inappropriate value for a correctly identified bond, or ignoring the C-O bond during the reaction.

(b) Reference to the effect of increasing pressure on the amount of ethanol produced at equilibrium was not always included. Many explanations were given in terms of the change producing a shift to the position of equilibrium to the right; the significance of producing a smaller amount of gas to explain this change was mentioned less frequently.

(c) (i) Many candidates were able to deduce the behaviour of H_3PO_4 as an acid or an electrophile. Some explanations were not specific to step 1 in the reaction.

(ii) Some candidates were able to identify the correct species in step 2. Explanations were often vague, for example, 'seeking electrons' or 'being attacked by a pair of electrons' rather than 'accepting a lone pair'.

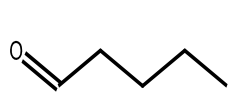
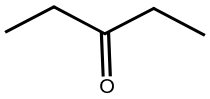
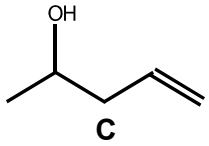
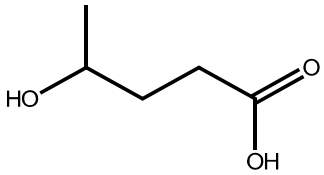
(iii) A small proportion of candidates applied knowledge of the drawing of mechanism arrows to represent electron movement when bonds are made and broken in this unfamiliar context. A common error showed arrows pointing in the opposite direction to the way an electron pair moves when a bond is made or broken. Sometimes an arrow did not start from the bond being broken or from the lone pair of electrons on an atom that is used to make a specific bond.

(iv) It was generally well known how a catalyst affects a reaction. Some ambiguity was seen in answers that described the effect of catalysts as increasing the number of collisions between particles rather than increasing the frequency of successful collisions.

(v) Stronger candidates clearly described the regeneration of the catalyst. Some candidates were vague and imprecise, for example 'the catalyst did not change in the reaction'.

(vi) Some candidates showed an excellent understanding of the consequence of more alkyl groups present on the C^+ of the intermediate in terms of a larger inductive effect. Some candidates incorrectly referred to the inductive effect of propan-2-ol rather than the carbocation intermediate which was responsible for the production of propan-2-ol.

(d) The sideways overlap of two p orbitals to form a pi bond between the two carbon atoms was well known. Descriptions of the formation of the sigma bond by direct overlap of sp^2 hybrid orbitals were less common.

5(a)	molecules / isomers with the same molecular formula / same number of atoms of each element **AND** different structural formulae / different structures	**1**
5(b)(i)	carbonyl (group)	**1**
5(b)(ii)	 **A**  **B**	**2**
5(c)(i)	 **C** **M1** $CH_3CH(OH)$ present **M2** five carbons in a straight chain **AND** molecular formula $C_5H_{10}O$ **AND** 1 C=C structure which shows only one type of stereoisomerism	**2**
5(c)(ii)	optical	**1**
5(d)(i)	ester	**1**
5(d)(ii)	condensation	**1**
5(d)(iii)		**1**

Question 5

(a) This definition was generally well known.

(b) (i) Many candidates appreciated that the two molecules were members of the carbonyl functional group. Some candidates incorrectly referred to them as either aldehydes or ketones.

(ii) Correct structures of **A** were common. Many candidates represented **B** as a ketone; the correct carbon skeleton was not always deduced.

(c) (i) It was common for some of the information in the question to be used to produce partially correct structures. Many candidates incorrectly produced structures with two types of stereoisomerism.

(ii) This was generally well answered.

(d) (i) Ketone and carboxylic acid were the most common incorrect answers.

(ii) Hydrolysis was a common incorrect response.

(iii) Deduction of the structure of **D** proved to be demanding. Often the structure drawn had a carboxylic acid group at one end of the molecule; identification of $(CH_3)CH(OH)$– at one end of the molecule was seen less frequently. Occasionally, the correct ends of the molecule were drawn with an incorrect number of carbon atoms in the carbon backbone.

(iv) Stronger candidates clearly identified two differences between the spectra in terms of the different absorption ranges produced as a result of the different environment of the C=O bond in an ester compared to a carboxyl group, or in terms of the presence of the O-H bond in a hydroxyl or carboxyl group which is absent in the ester, **E**. Reference to absorbance at 2850 – 2950 due to C-H bonds was common, even though both **D** and **E** contain this type of bond.

1(a)	<table><tr><td></td><td>nucleon number</td><td>number of neutrons</td><td>number of electrons</td></tr><tr><td>Tellurium-130</td><td>130</td><td>78</td><td>52</td></tr></table>	**3**
1(f)(i)	$TeF_x + xH_2O \rightarrow Te(OH)_x + xHF$ **OR** $TeF_6 + 6H_2O \rightarrow Te(OH)_6 + 6HF$	**1**
1f(ii)	hydrolysis	**1**

Question 1

(a) This was generally well answered. Some candidates were not familiar with the term nucleon number, often interpreting this as neutron number.

(b) Many candidates answered this question correctly. Common errors included not specifying **1**s and giving 5p as the answer.

(c) Candidates generally answered this well. State symbols were not always included, especially in Te^{+}.

(d) (i) The majority of candidates correctly identified two factors. Many candidates incorrectly stated that the shielding effect remains constant. An 'increase in nuclear charge' was a common incorrect response that negated other descriptions.

(ii) Most candidates recognised a general increase in successive ionisation energies, with a majority of those able to indicate the most significant jump from the 6th to 7th ionisation energy. Many answers incorrectly included a graph with a dip, most often in energy at the $6/7^{th}$ ionisation energy.

(e) Candidates who correctly referenced spin-pair interactions gained credit. Few candidates could use the VSEPR theory correctly.

(f) (i) This was generally not answered well. Many responses included a mixture of numbers and '*x*'s. A small proportion of candidates gave a correctly balanced equation.

(ii) Many candidates correctly identified this as a 'hydrolysis' reaction. A common answer was 'displacement', which was not enough to gain credit.

3(a)	**M1** *appearance* (colourless / yellow) liquid **M2** *structure and bonding* simple / molecular **AND** covalent	**2**

3(d)(i)	ultraviolet light	**1**

3(d)(ii)

name of stage	equation	
initiation	$Cl_2 \rightarrow 2Cl\bullet$	[1]
propagation	$CH_2Cl2 + Cl\bullet \rightarrow CHCl_2\bullet + HCl$	[1]
termination	$CHCl_2\bullet + Cl\bullet \rightarrow CHCl_3$	[1]

3

Question 3

(a) Very few candidates could state the appearance of $SiCl_4$ at room temperature; more were able to state the correct structure and bonding. The most common incorrect response was giant covalent.

(b) Most candidates correctly referred to the appearance and pH. Candidates often did not refer to the relevant observation and struggled to differentiate the behaviours of the two chlorides.

(c) (i) A large number of candidates stated the correct reagent. Many left out aqueous or dilute as the conditions.

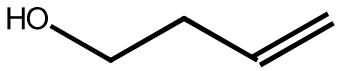
(ii) Many of the candidates defined disproportionation; a small proportion were able to refer to the correct oxidation numbers. Several referred incorrectly to the +1 oxidation number for chlorine.

(d) (i) This was well answered by most candidates.

(ii) Most candidates correctly answered the initiation step. Many candidates found the propagation and termination steps more challenging.

(e) This was well answered by most students.

(f) Many candidates were able to give the correct formulae at m/e = 164.Very few candidates gave both of the correct options at m/e = 166.

4(a)(i)		**1**
4(a)(ii)	reducing agent	**1**
4(b)(i)	cold dilute acidified $KMnO_4$	**1**
4(b)(ii)	CH_2O	**1**
4(b)(iii)	<table><tr><td>type of hybridisation</td><td>sp^2</td><td>sp^3</td></tr><tr><td>number of carbon atoms in **V**</td><td>3</td><td>1</td></tr></table>	**2**

Question 4

(a) (i) Common errors were to retain the COOH group and to have the C=C double bond reduced.

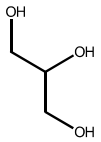
(ii) This was well answered by most candidates.

(b) (i) This was generally well answered, although few candidates stated all the required conditions.

(ii) This was well answered by most candidates.

(iii) A small proportion of candidates gained full credit here.

(c) Many candidates correctly gave pentan-1-ol. A small number of candidates made the link between the yellow precipitate with alkaline I_2(aq) and a $CH_3CH(OH)$– group. Many candidates struggled to synthesise all the detail to present a cogent answer to the question.

5(a)(i)	$C_8H_{18} + 12½\ O_2 \rightarrow 8CO_2 + 9H_2O$ **M1** correct formula of M C_8H_{18} **M2** balanced equation showing complete combustion of M	**2**
5(c)(i)	*suggest suitable **reagent** and conditions for step 1* heat **AND** any dilute named mineral acid e.g. dilute sulfuric acid / dilute hydrochloric acid	**1**
5(c)(ii)	 **OR** $CH_2OHCH(OH)CH_2OH$	**1**
5(c)(iii)	condensation	**1**
5(c)(iv)	methanol	**1**
5(c)(v)	methyl decanoate	**1**

Question 5

(a) (i) This was well answered by most candidates, although balancing the equation proved challenging for some.

(ii) Most candidates correctly mentioned the production of CO instead of CO_2. A very small number correctly referred to C and unburnt hydrocarbons.

(b) (i) A large proportion of candidates incorrectly interpreted this as a question about catalytic converters. Many answers used NO instead of N_2.

(ii) A small number of candidates correctly identified the catalytic role of NO_x and gave all the relevant equations. Many gave the equation for the formation of H_2SO_4, and a significant number gave equations and answers related to the formation of HNO_3 for acid rain.

(iii) This was well answered by most. Some candidates gave the identity of a gas instead of the type of air pollution.

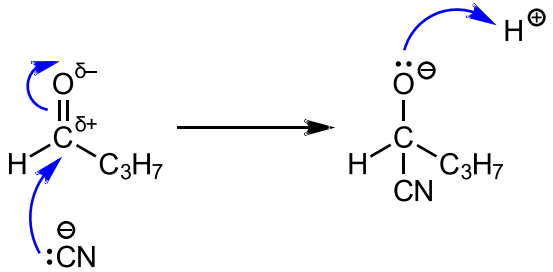
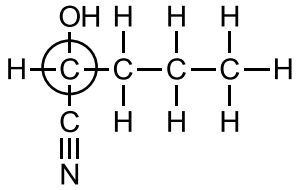
(c) (i) Candidates often omitted dilute/(aq) with a named acid here.

(ii) A few candidates recognised that all three ester groups will undergo hydrolysis to form propanetriol. Many answers gave fragments where only one ester group had undergone hydrolysis.

(iii) This was well answered by most candidates. The majority of answers referred to esterification/substitution instead of condensation.

(iv) A small proportion of candidates answered this question correctly.

(v) A small proportion of candidates answered this question correctly.

3(a)(i)	(a reaction where) two (or more) compounds / reagents / molecules form (only) **one product**	**1**
3(a)(ii)	position(al isomerism)	**1**
3(a)(iii)	butanal	**1**
3(b)(i)	 curly arrow from lone pair on C of CN^- to $C^{(\delta+)}$	**1**
	correct dipole on C=O **AND** curly arrow from C=O bond to $O^{\delta-}$	**1**
	correct organic intermediate	**1**
	curly arrow from lone pair on alkoxide $O^{(-)}$ to H^+ / H of HCN / H of H_2O (to reform the alcohol)	**1**
3(b)(iv)		**1**

Question 3

This question was largely framed around organic chemistry, with the inclusion of some physical and stoichiometric work. It was well attempted on the whole, although knowledge of some key definitions was seen to be weaker.

(a) (i) Many candidates appreciated that a single product was the outcome of an addition reaction, but many others described/attempted to describe an addition reaction without indicating that only one product was formed.

(ii) This was well answered by most candidates.

(iii) The major errors here were butan-1-one, butan-1-ol/butanol or propanal/propan-1-al. Candidates are reminded that there is no need to offer a locational number for the aldehyde group.

(iv) Very few candidates converted the quantities to moles, so simply offered 96% of 5000 kg as the answer.

(b) (i) Candidates seemed largely unaware of the nucleophilic addition mechanism. Common errors and omissions included:

- incorrect placement of the lone pair of electrons on the N of the CN^- ion.
- giving the intermediate a positive charge which was subsequently attacked by the CN^- nucleophile.
- mechanisms involving the H (of HCN) / H^+ attacking the intermediate negative ions.

- curly arrows inaccurately coming from the C (in the C=O group).
- the H (of HCN) shown attacking the O (in the C=O group) to form an intermediate hydroxyl derivative with a positive charge.

(ii) The first and third marks of this question gave the largest number of variants, with Fehling's reagent or DNPH offered for experiment 1. For experiment 3, many answers erroneously gave a change or loss of colour rather than a method of noting that no reaction had occurred.

(iii) There were some very good answers here, although some increased the number of H atoms in the product and/or H_2 appeared as a product.

(iv) This was generally very well answered, with occasionally some candidates circling two C atoms or the entirety of HOCHCN.

(c) It appeared that many candidates did not understand the term molecular formula and gave structural formulae throughout.

(d) (i) Some answers referred to a change in position of the equilibrium/yield, which was not credited.

(ii) Many answers excluded the word gas and just stated 'less/fewer (number of) moles'. Although the reaction in question was entirely in the gas phase, precision in an answer is crucial. A few candidates gave the incorrect direction of the equilibrium shift.

(iii) Many candidates used a value of 436 in their answers, which suggests the use of a Data Booklet. Centres are reminded that Data Booklets are **not** to be issued to candidates for these papers.

(iv) Any mention of hydrogen bonding was not credited, in accordance with the convention and definition used by this qualification.

4(a)(i)	NaOH in alcohol / ethanol AND heat (under reflux)	**1**

4(c)(i)	substitution	**1**

4(c)(ii)	reducing agent	**1**

Question 4

This was a second organic chemistry question, which included some analytical work. This was generally answered well by many candidates, though later parts on IR and MS appeared less familiar to some.

(a) (i) 'Heat' or equivalent was often missing. H_2SO_4 was a popular answer despite the conditions suggested in the syllabus.

(ii) Many repeat units were shown containing four carbon atoms, as if the CH_3 groups had been incorporated into the backbone. Some candidates showed an ethene structure with missing CH_3 side groups.

(iii) Candidates very often showed the reverse equation with C_4H_8 and H_2O. Quite a few answers contained Al_2O_3 as a reactant and several answers omitted H_2O as a product.

(b) (i) This was well answered.

(ii) There were many correct structures/names for iodoform but very few correct propanoate ions.

(c) (i) There were many correct answers to this item.

(ii) Catalyst and oxidising agent were the common errors here, but otherwise there were many correct answers.

(d) (i) Compound **F** was commonly given incorrectly as the unknown compound, referring to an O–H absorption in the range 2850–3000 cm^{-1}. Candidates are reminded that C–H stretches appear in all organic spectra. Where absorptions are referenced, candidates need to identify which bond is stretching: 'nitrile', for example, is insufficient.

(ii) There were many null responses to this. Where attempted, candidates' answers were evenly split between correct identification of an identical abundance and some complicated (erroneous) mathematics.